Joshua Li

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EDUCATION

Arizona State University | Tempe, AZ

Masters of Science, Computer Science

Exp. Graduate: Dec 2026, GPA: 4.0

University of California, San Diego | San Diego, CA

Bachelor of Science, Data Science

Graduated: Mar 2025, GPA: 3.9

WORK EXPERIENCE

BILL | San Jose, CA

Associate Fraud Risk Strategy Data Scientist | Dec 2025 - Present

- Engineered a multi-objective NSGA-II genetic algorithm via Optuna to jointly optimize 15 production rules, processing 15,000 trials in 2 minutes and reducing manual referrals by 4.4% per quarter.
- Prototyped a KYC fraud detection system combining a Random Forest model (0.79 AUC), a NetworkX identity graph utilizing vectorized identity signals, and an agentic AI LiteLLM tool-calling loop to synthesize investigation tickets.
- Deployed an end-to-end AWS SageMaker batch ML pipeline utilizing graph-based fanout, achieving 99.5% recall and 94.5% precision on reachable fraud.
- Identified and resolved a portfolio-wide pipeline bug affecting 411,000 payment events across a 6 month window, enabling clean rule re-tuning.
- Developed automated KPI dashboards with Tableau, providing stakeholders with real-time visibility into fraud prevention trends, rule performance week-over-week, and loss prevention metrics.

Single Particle LLC | San Diego, CA

Software Programming Intern | Jul 2023 - Oct 2023

- Fine-tuned ChatGPT models for cryo-electron microscopy (cryoEM) workstations and led the integration of AI into the company website.

Intel Corporation | Folsom, CA

Software Programming Intern | Jun 2022 - Sep 2022

- Engineered and rigorously tested data science workflows tailored to various industry sectors, enhancing team insights into workstation performance.
- Conducted extensive benchmarking of workflows by simulating real-world industry use cases using Python and Scikit-Learn, improving efficiency by 20%.

PROJECTS

Optimizing Training Cost for Scalable Graph Processing | UCSD, Qualcomm

Python, PyTorch, NetworkX, CUDA

- Worked closely with Qualcomm to develop a scalable graph-based learning pipeline to predict congestion in chip designs using the DE-HNN model.
- Processed 6 large-scale netlists (460k to 920k nodes) into bipartite graph structures, integrating spectral and structural features for demand prediction.
- Achieved 89.3% runtime reduction and 38.8% peak GPU memory savings.
- Delivered a final model with only a 6% average performance drop while dramatically lowering compute costs.

SKILLS

- **Languages**: Python, SQL, R, C, C++, Java, JavaScript, TypeScript, HTML/CSS, MATLAB
- **Frameworks & Libraries**: Pandas, PyTorch, TensorFlow, Keras, Dask, Scikit-learn, NetworkX, Plotly, Seaborn, React, D3, Flask, Django, Spark, Hadoop, Optuna
- **Databases & Big Data**: MySQL, Postgres, NoSQL, MongoDB, Starburst, Databricks
- **Tools & Platforms**: AWS (Sagemaker, S3), Google Cloud, Docker, Tableau, Git, GitHub, JSON, Microsoft Excel, Node.js, Data Visor, LiteLLM, Ollama